

Dahye Kim

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RESEARCH INTERESTS

Image and Video Generation; Multimodal LLMs; Model Efficiency; Interpretability

EDUCATION

Boston University Ph.D. in Computer Science Advisor: Prof. Deepti Ghadiyaram	Boston, MA, USA Sep. 2024 – Present
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Yonsei University M.S. in Electrical and Electronic Engineering Advisor: Prof. Kwanghoon Sohn	Seoul, South Korea Sep. 2022 – Aug. 2024
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B.S. in Electrical and Electronic Engineering GPA: 4.0/4.0; class rank: 2/224	Mar. 2020 – Aug. 2022
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Hongik University Electronic and Electrical Engineering	Seoul, South Korea Mar. 2018 – Feb. 2020
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PUBLICATIONS

- [1] **Swift Sampling: Selecting Temporal Surprises via Taylor Series.**
Dahye Kim, Bhuvan Sachdeva, Karan Uppal, Naman Gupta,
Vineeth N. Balasubramanian, and Deepti Ghadiyaram.
NeurIPS, 2026. Accepted. [\[PDF\]](#) [\[Project Page\]](#)
- [2] **DDiT: Dynamic Patch Scheduling for Efficient Diffusion Transformers.**
Dahye Kim, Deepti Ghadiyaram, and Raghudeep Gadde.
CVPR, 2026. **Highlight.** [\[PDF\]](#) [\[Project Page\]](#)
- [3] **Revelio: Interpreting and leveraging visual semantic information in diffusion models.**
Dahye Kim*, Xavier Thomas*, and Deepti Ghadiyaram.
ICCV, 2025; MIV Workshop, CVPR 2025 (**Oral**). [\[PDF\]](#) [\[Project Page\]](#)
- [4] **Language-free Training for Zero-shot Video Grounding.**
Dahye Kim, Jungin Park, Jiyoung Lee, Seongheon Park, and Kwanghoon Sohn.
WACV, 2023. **Oral.** [\[PDF\]](#)
- [5] **Normality Guided Multiple Instance Learning for Weakly Supervised Video Anomaly Detection.**
Seongheon Park, Hanjae Kim, Minsu Kim, Dahye Kim, and Kwanghoon Sohn.
WACV, 2023. **Oral.** [\[PDF\]](#)

*Equal contribution.

PREPRINTS

Concept Steerers: Leveraging K-Sparse Autoencoders for Test-Time Controllable Generations.
Dahye Kim and Deepti Ghadiyaram.
arXiv preprint, 2025. [\[PDF\]](#) [\[Project Page\]](#)

RESEARCH EXPERIENCE

Adobe Inc. <i>Research Scientist Intern (Mentor: Connelly Barnes)</i> Diffusion models	Seattle, WA, USA Aug. 2026 – Present
Boston University <i>Research Assistant (Advisor: Prof. Deepti Ghadiyaram)</i> Generative models, multimodal learning, and interpretability	Boston, MA, USA Sep. 2024 – Present
Amazon.com, Inc. <i>Applied Scientist Intern, GenAI Team (Mentor: Raghudeep Gadde)</i> Efficient diffusion models: DDiT (CVPR 2026, Highlight)	Seattle, WA, USA Jun. 2025 – Nov. 2025
Digital Image Media Lab., Yonsei University Advisor: Prof. Kwanghoon Sohn <i>Research Assistant</i> Multimodal learning with visual and textual representations	Seoul, South Korea Sep. 2022 – Jul. 2024
<i>Undergraduate Research Assistant</i> Human-object interaction detection and video understanding	Jan. 2021 – Aug. 2022

AWARDS AND HONORS

Hariri Institute Student Excellence Award , Boston University <i>Awarded for significant contributions within the Hariri Institute and a commitment to excellence.</i>	2026
Brain Korea 21 (BK21) Scholarship, National Research Foundation of Korea	2022 – 2024
Automotive System IC Scholarship	2022 – 2024
Teaching Assistant Scholarship, Yonsei University	Fall 2022, Fall 2023
Graduate Student Research Assistant Scholarship, Yonsei University	Fall 2022
High Honors Student, top 3% of department	Fall 2020; Spring, Fall 2021
Academic Excellence Scholarship, Yonsei University	Spring, Fall 2021
Honors Student, top 10% of department	Spring 2020
National Science and Technology Scholarship	Fall 2020
Academic Excellence Scholarship, Hongik University	Fall 2018; Spring, Fall 2019

PROFESSIONAL SERVICE

<i>Workshop Organizer</i> XAI4CV: Explainable Artificial Intelligence for Computer Vision	CVPR 2025, CVPR 2026
<i>Student Volunteer</i> 3rd New England Mechanistic Interpretability (NEMI) Workshop, Boston University	Aug. 2026
<i>Reviewer</i> Conferences: CVPR (2026), ECCV (2026), NeurIPS (2026), AAAI (2027) Journals: IJCV Workshops: CVPR — MIV (2025), AI4CC (2026)	

TEACHING EXPERIENCE

CS 541: Applied Machine Learning	Spring 2025
EEE 3313: Introductory Digital Labs	Fall 2023
EEE 2060: Signals and Systems	Fall 2022

SKILLS AND LANGUAGES

Programming and tools: Python, MATLAB, C/C++, PyTorch, Linux shell, $\LaTeX$.
Languages: Korean (native), English (fluent).